

# A/B TESTING ANALYSIS PROJECT IN PYTHON

## IMPORT PYTHON LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind

pd.set_option('display.max_columns', None)
sns.set_style("whitegrid")
```

## LOAD DATA

```
control_df = pd.read_csv("control_group.csv", sep=';')
test_df = pd.read_csv("test_group.csv", sep=';')
```

## BASIC INSPECTION

```
control_df.shape
test_df.shape
```

```
(30, 10)
```

Both Control and Test Campaign has 30 rows and 10 columns respectively

## FIRST FIVE ROWS OF EACH CAMPAIGN

```
control_df.head()

{"summary": "{\n  \"name\": \"control_df\",\n  \"rows\": 30,\n  \"fields\": [\n    {\n      \"column\": \"Campaign Name\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 1,\n        \"samples\": [\n          \"Control Campaign\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Date\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 30,\n        \"samples\": [\n
```

```

\"28.08.2019\"\\n        ],\\n        \"semantic_type\\\": \"\\\",\\n
\"description\\\": \"\\\"\\n        }\\n        },\\n        {\\n        \"column\\\":
\"Spend [USD]\\\",\\n        \"properties\\\": {\\n        \"dtype\\\":
\"number\\\",\\n        \"std\\\": 367,\\n        \"min\\\": 1757,\\n
\"max\\\": 3083,\\n        \"num_unique_values\\\": 30,\\n
\"samples\\\": [\\n        2421\\n        ],\\n        \"semantic_type\\\":
\"\\\",\\n        \"description\\\": \"\\\"\\n        }\\n        },\\n        {\\n
\"column\\\": \"# of Impressions\\\",\\n        \"properties\\\": {\\n
\"dtype\\\": \"number\\\",\\n        \"std\\\": 21688.922907879514,\\n
\"min\\\": 71274.0,\\n        \"max\\\": 145248.0,\\n
\"num_unique_values\\\": 29,\\n        \"samples\\\": [\\n        92029.0\\n
        ],\\n        \"semantic_type\\\": \"\\\",\\n
\"description\\\": \"\\\"\\n        }\\n        },\\n        {\\n        \"column\\\":
\"Reach\\\",\\n        \"properties\\\": {\\n        \"dtype\\\": \"number\\\",\\n
\"std\\\": 21832.349595318796,\\n        \"min\\\": 42859.0,\\n
\"max\\\": 127852.0,\\n        \"num_unique_values\\\": 29,\\n
\"samples\\\": [\\n        74192.0\\n        ],\\n
\"semantic_type\\\": \"\\\",\\n        \"description\\\": \"\\\"\\n        }\\n
        },\\n        {\\n        \"column\\\": \"# of Website Clicks\\\",\\n
\"properties\\\": {\\n        \"dtype\\\": \"number\\\",\\n        \"std\\\":
1757.369003029211,\\n        \"min\\\": 2277.0,\\n        \"max\\\":
8137.0,\\n        \"num_unique_values\\\": 29,\\n        \"samples\\\": [\\n
8127.0\\n        ],\\n        \"semantic_type\\\": \"\\\",\\n
\"description\\\": \"\\\"\\n        }\\n        },\\n        {\\n        \"column\\\": \"#
of Searches\\\",\\n        \"properties\\\": {\\n        \"dtype\\\":
\"number\\\",\\n        \"std\\\": 866.0893678503669,\\n        \"min\\\":
1001.0,\\n        \"max\\\": 4891.0,\\n        \"num_unique_values\\\": 29,\\n
        \"samples\\\": [\\n        4891.0\\n        ],\\n
\"semantic_type\\\": \"\\\",\\n        \"description\\\": \"\\\"\\n        }\\n
        },\\n        {\\n        \"column\\\": \"# of View Content\\\",\\n
\"properties\\\": {\\n        \"dtype\\\": \"number\\\",\\n        \"std\\\":
777.5454685515341,\\n        \"min\\\": 848.0,\\n        \"max\\\": 4219.0,\\n
        \"num_unique_values\\\": 26,\\n        \"samples\\\": [\\n
1984.0\\n        ],\\n        \"semantic_type\\\": \"\\\",\\n
\"description\\\": \"\\\"\\n        }\\n        },\\n        {\\n        \"column\\\": \"#
of Add to Cart\\\",\\n        \"properties\\\": {\\n        \"dtype\\\":
\"number\\\",\\n        \"std\\\": 407.45797329295203,\\n        \"min\\\":
442.0,\\n        \"max\\\": 1913.0,\\n        \"num_unique_values\\\": 29,\\n
\"samples\\\": [\\n        1486.0\\n        ],\\n
\"semantic_type\\\": \"\\\",\\n        \"description\\\": \"\\\"\\n        }\\n
        },\\n        {\\n        \"column\\\": \"# of Purchase\\\",\\n
\"properties\\\": {\\n        \"dtype\\\": \"number\\\",\\n        \"std\\\":
185.02864243723857,\\n        \"min\\\": 222.0,\\n        \"max\\\": 800.0,\\n
        \"num_unique_values\\\": 28,\\n        \"samples\\\": [\\n
475.0\\n        ],\\n        \"semantic_type\\\": \"\\\",\\n
\"description\\\": \"\\\"\\n        }\\n        }\\n    ]\\n
n}\\\", \"type\": \"dataframe\", \"variable_name\": \"control_df\"}

```

```
test_df.head()
```

```

{"summary":{"\n  \"name\": \"test_df\",\n  \"rows\": 30,\n  \"fields\": [\n    {\n      \"column\": \"Campaign Name\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 1,\n        \"samples\": [\n          \"Test Campaign\",\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Date\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 30,\n        \"samples\": [\n          \"28.08.2019\",\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Spend [USD]\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 348,\n        \"min\": 1968,\n        \"max\": 3112,\n        \"num_unique_values\": 30,\n        \"samples\": [\n          2247,\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Impressions\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 32121,\n        \"min\": 22521,\n        \"max\": 133771,\n        \"num_unique_values\": 30,\n        \"samples\": [\n          54627,\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Reach\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 28795,\n        \"min\": 10598,\n        \"max\": 109834,\n        \"num_unique_values\": 30,\n        \"samples\": [\n          41267,\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Website Clicks\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1708,\n        \"min\": 3038,\n        \"max\": 8264,\n        \"num_unique_values\": 30,\n        \"samples\": [\n          8144,\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Searches\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 388,\n        \"min\": 1854,\n        \"max\": 2978,\n        \"num_unique_values\": 29,\n        \"samples\": [\n          1920,\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of View Content\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 597,\n        \"min\": 858,\n        \"max\": 2801,\n        \"num_unique_values\": 30,\n        \"samples\": [\n          1281,\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Add to Cart\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 347,\n        \"min\": 278,\n        \"max\": 1391,\n        \"num_unique_values\": 30,\n        \"samples\": [\n          1009,\n          ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Purchase\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 211,\n        \"min\": 238,\n        \"max\": 890,\n        \"num_unique_values\": 28,\n        \"samples\": [\n          275,\n          ]\n        ]\n      }\n    }\n  ]\n}

```

```
],\n    \"semantic_type\": \"\",\n    \"description\": \"\"\n}\n]\n}","type":"dataframe","variable_name":"test_df"}
```

## MISSING VALUES

```
control_df.isnull().sum()
```

|                     |   |
|---------------------|---|
| Campaign Name       | 0 |
| Date                | 0 |
| Spend [USD]         | 0 |
| # of Impressions    | 1 |
| Reach               | 1 |
| # of Website Clicks | 1 |
| # of Searches       | 1 |
| # of View Content   | 1 |
| # of Add to Cart    | 1 |
| # of Purchase       | 1 |

dtype: int64

```
test_df.isnull().sum()
```

|                     |   |
|---------------------|---|
| Campaign Name       | 0 |
| Date                | 0 |
| Spend [USD]         | 0 |
| # of Impressions    | 0 |
| Reach               | 0 |
| # of Website Clicks | 0 |
| # of Searches       | 0 |
| # of View Content   | 0 |
| # of Add to Cart    | 0 |
| # of Purchase       | 0 |

dtype: int64

# INSIGHTS

Both campaigns originally have 30 daily observations

The Control campaign contains one incomplete record, so we must clean it before analysis

The Test campaign is complete and ready to use

## DATA CLEANING

1) Since only one row in the Control campaign has missing values, we will drop that row.

2) This is acceptable because only one observation is affected.

3) We also convert the Date column into proper datetime format.

```
control_df = control_df.dropna()
test_df = test_df.dropna()

control_df['Date'] = pd.to_datetime(control_df['Date'], format='%d.%m.%Y')
test_df['Date'] = pd.to_datetime(test_df['Date'], format='%d.%m.%Y')
```

## Add Campaign Labels and Merge Both Datasets

To compare both campaigns together, we add a new column called Group and then merge the two datasets.

```
control_df['Group'] = 'Control'
test_df['Group'] = 'Test'

ab_data = pd.concat([control_df, test_df], ignore_index=True)

control_df.shape
test_df.shape
```

```
(30, 10)
```

```
ab_data.shape
```

```
(59, 11)
```

## Create KPI Metrics

Raw numbers like impressions and clicks are useful, but KPI ratios help us evaluate the campaign more accurately.

We calculate:

CTR = Click Through Rate

Conversion Rate

CPC = Cost Per Click

CPA = Cost Per Acquisition / Purchase

Add to Cart Rate

Purchase from Cart Rate

```
ab_data['CTR'] = (ab_data['# of Website Clicks'] / ab_data['# of  
Impressions']) * 100  
ab_data['Conversion Rate'] = (ab_data['# of Purchase'] / ab_data['# of  
Website Clicks']) * 100  
ab_data['CPC'] = ab_data['Spend [USD]'] / ab_data['# of Website  
Clicks']  
ab_data['CPA'] = ab_data['Spend [USD]'] / ab_data['# of Purchase']  
ab_data['Add to Cart Rate'] = (ab_data['# of Add to Cart'] /  
ab_data['# of View Content']) * 100  
ab_data['Purchase from Cart Rate'] = (ab_data['# of Purchase'] /  
ab_data['# of Add to Cart']) * 100
```

## STATISTICS VALUE

```
ab_data.describe()
```

```

{"summary":{"\n  \"name\": \"ab_data\",\n  \"rows\": 8,\n  \"fields\": [\n    {\n      \"column\": \"Date\",\n      \"properties\": {\n        \"dtype\": \"date\",\n        \"min\": \"1970-01-01 00:00:00.000000059\",\n        \"max\": \"2019-08-30 00:00:00\",\n        \"num_unique_values\": 7,\n        \"samples\": [\n          \"59\",\n          \"2019-08-15 16:16:16.271186432\",\n          \"2019-08-23 00:00:00\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Spend [USD]\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1102.640513563197,\n        \"min\": 59.0,\n        \"max\": 3112.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          2435.762711864407,\n          2743.0,\n          59.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Impressions\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 50106.204780276894,\n        \"min\": 59.0,\n        \"max\": 145248.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          91775.8813559322,\n          117315.0,\n          59.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Reach\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 43626.63993928129,\n        \"min\": 59.0,\n        \"max\": 127852.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          70868.64406779662,\n          95490.5,\n          59.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Website Clicks\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2842.645913494397,\n        \"min\": 59.0,\n        \"max\": 8264.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          5682.593220338983,\n          7218.5,\n          59.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Searches\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1500.3468641752884,\n        \"min\": 59.0,\n        \"max\": 4891.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          2321.813559322034,\n          2778.5,\n          59.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of View Content\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1288.7389234691066,\n        \"min\": 59.0,\n        \"max\": 4219.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          1900.1694915254238,\n          2424.0,\n          59.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Add to Cart\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 614.6048678292085,\n        \"min\": 59.0,\n        \"max\": 1913.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          1087.2203389830509,\n          1386.5,\n          59.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"# of Purchase\",

```

```

{"properties": {"dtype": "number", "std": 277.1147391950759, "min": 59.0, "max": 890.0, "num_unique_values": 8, "samples": [693.0, 59.0]}, "semantic_type": "", "description": "", "column": "CTR", "properties": {"dtype": "number", "std": 20.10567480829817, "min": 1.8575459285266986, "max": 59.0, "num_unique_values": 8, "samples": [7.7126787355586846, 8.452497421597295, 59.0]}, "semantic_type": "", "description": "", "column": "Conversion Rate", "properties": {"dtype": "number", "std": 19.113100286894863, "min": 3.0088495575221237, "max": 59.0, "num_unique_values": 8, "samples": [10.335158384315717, 13.051025621510044, 59.0]}, "semantic_type": "", "description": "", "column": "CPC", "properties": {"dtype": "number", "std": 20.69570167272279, "min": 0.19924578547958086, "max": 59.0, "num_unique_values": 8, "samples": [0.48159090839461405, 0.5868956114151629, 59.0]}, "semantic_type": "", "description": "", "column": "CPA", "properties": {"dtype": "number", "std": 19.202471257206202, "min": 2.25375, "max": 59.0, "num_unique_values": 8, "samples": [5.483144124526937, 7.5204998392799745, 59.0]}, "semantic_type": "", "description": "", "column": "Add to Cart Rate", "properties": {"dtype": "number", "std": 57.459882568278736, "min": 13.473423980222496, "max": 201.5330188679245, "num_unique_values": 8, "samples": [64.42478954146058, 81.12434940517045, 59.0]}, "semantic_type": "", "description": "", "column": "Purchase from Cart Rate", "properties": {"dtype": "number", "std": 41.97308673967764, "min": 14.33628318584071, "max": 151.5837104072398, "num_unique_values": 8, "samples": [53.89260419591541, 66.69176775381686, 59.0]}, "semantic_type": "", "description": ""}
], "type": "dataframe"}

```



## GROUP WISE ANALYSIS

```
group_avg = ab_data.groupby('Group').mean(numeric_only=True).round(2)
print(group_avg)
```

|         | Spend [USD] | # of Impressions | Reach    | # of Website Clicks |
|---------|-------------|------------------|----------|---------------------|
| \ Group |             |                  |          |                     |
| Control | 2304.07     | 109559.76        | 88844.93 | 5320.79             |
| Test    | 2563.07     | 74584.80         | 53491.57 | 6032.33             |

|         | # of Searches | # of View Content | # of Add to Cart | # of Purchase |
|---------|---------------|-------------------|------------------|---------------|
| \ Group |               |                   |                  |               |
| Control | 2221.31       | 1943.79           | 1300.00          | 522.79        |
| Test    | 2418.97       | 1858.00           | 881.53           | 521.23        |

|         | CTR   | Conversion Rate | CPC  | CPA  | Add to Cart Rate | \     |
|---------|-------|-----------------|------|------|------------------|-------|
| Group   |       |                 |      |      |                  |       |
| Control | 5.10  | 11.48           | 0.49 | 5.05 |                  | 77.79 |
| Test    | 10.24 | 9.23            | 0.47 | 5.90 |                  | 51.51 |

|         | Purchase from Cart Rate |
|---------|-------------------------|
| Group   |                         |
| Control | 45.72                   |
| Test    | 61.79                   |

## MAIN COMPARISON-Average Metrics by Campaign

```
comparison = ab_data.groupby('Group')[[
    'Spend [USD]',
    '# of Impressions',
    'Reach',
    '# of Website Clicks',
    '# of Searches',
    '# of View Content',
    '# of Add to Cart',
    '# of Purchase'
]].mean().round(2)
print(comparison)
```

| \ Group        | Spend [USD]   | # of Impressions  | Reach            | # of Website Clicks |
|----------------|---------------|-------------------|------------------|---------------------|
| Control        | 2304.07       | 109559.76         | 88844.93         | 5320.79             |
| Test           | 2563.07       | 74584.80          | 53491.57         | 6032.33             |
| Purchase Group | # of Searches | # of View Content | # of Add to Cart | # of                |
| Control        | 2221.31       | 1943.79           | 1300.00          | 522.79              |
| Test           | 2418.97       | 1858.00           | 881.53           | 521.23              |

### 1. Test campaign spent more money

Control Spend: 2304.07

Test Spend: 2563.07

So the Test campaign is more expensive on average.

### 2. Control campaign generated much higher impressions and reach

Impressions: Control = 109,559.76 vs Test = 74,584.80

Reach: Control = 88,844.93 vs Test = 53,491.57

This means the Control campaign reached a much larger audience.

### 3. Test campaign generated more website clicks

Control Clicks: 5320.79

Test Clicks: 6032.33

Even though Test reached fewer people, it got more clicks, which suggests better engagement quality.

### 4. Test campaign also generated more searches

Control Searches: 2221.31

Test Searches: 2418.97

This suggests users exposed to the Test campaign were more interested in exploring the product/brand.

### 5. Control campaign produced more Add-to-Cart actions

Control Add to Cart: 1300.00

Test Add to Cart: 881.53

This is a very important finding:

Although Test got more clicks, Control pushed more users deeper into the funnel.

6. Purchases are almost identical

Control Purchases: 522.79

Test Purchases: 521.23

So at the final purchase stage, both campaigns performed almost the same.

## COMPARING KPI METRICES

```
kpi_comparison = ab_data.groupby('Group')[[
    'CTR',
    'Conversion Rate',
    'CPC',
    'CPA',
    'Add to Cart Rate',
    'Purchase from Cart Rate'
]].mean().round(2)
(print(kpi_comparison))
```

|         | CTR   | Conversion Rate | CPC  | CPA  | Add to Cart Rate | \ |
|---------|-------|-----------------|------|------|------------------|---|
| Group   |       |                 |      |      |                  |   |
| Control | 5.10  | 11.48           | 0.49 | 5.05 | 77.79            |   |
| Test    | 10.24 | 9.23            | 0.47 | 5.90 | 51.51            |   |

|         | Purchase from Cart Rate |
|---------|-------------------------|
| Group   |                         |
| Control | 45.72                   |
| Test    | 61.79                   |

## A) CTR (Click Through Rate)

Actual Output

Control CTR: 5.10%

Test CTR: 10.24%

### Insight

This is one of the biggest findings in the project.

The Test campaign has nearly double the CTR of the Control campaign.

That means the Test campaign ad was much better at attracting clicks from people who saw it.

Business meaning

The Test campaign is stronger at top-of-funnel engagement.

## B) Conversion Rate

Actual Output

Control Conversion Rate: 11.48%

Test Conversion Rate: 9.23%

### Insight

Even though Test got more clicks, Control converted clicks into purchases better.

Business meaning

Control campaign visitors were more likely to purchase after clicking.

## C) CPC (Cost Per Click)

Actual Output

Control CPC: 0.49

Test CPC: 0.47

## Insight

The Test campaign got clicks at a slightly lower cost.

## Business meaning

From a click acquisition perspective, Test is a little more efficient.

## D) CPA (Cost Per Acquisition / Purchase)

Actual Output

Control CPA: 5.05

Test CPA: 5.90

## Insight

Control campaign acquires a purchase at a lower cost than the Test campaign.

## Business meaning

Even though Test is better at generating clicks, Control is more cost-efficient at generating actual purchases.

## E) Add to Cart Rate

### Actual Output

Control Add-to-Cart Rate: 77.79%

Test Add-to-Cart Rate: 51.51%

## Insight

This is another major finding.

A much larger proportion of Control users moved from viewing content → adding to cart.

## Business meaning

The Control campaign seems to attract more purchase-ready users, or the quality of traffic from Control is better.

## F) Purchase from Cart Rate

### Actual Output

Control: 45.72%

Test: 61.79%

## Insight

Although Test had fewer add-to-cart actions, users who did add to cart were more likely to complete the purchase.

## Business meaning

The bottom-funnel behavior is stronger in the Test campaign once the user reaches cart stage.

## Overall Interpretation

### Test Campaign

- 1) Higher CTR
- 2) More clicks
- 3) More searches
- 4) Slightly lower CPC

So Test is better at attracting attention and generating traffic.

### Control Campaign

- 1) Higher conversion rate
- 2) Higher add-to-cart rate
- 3) Lower CPA
- 4) Slightly more purchases overall



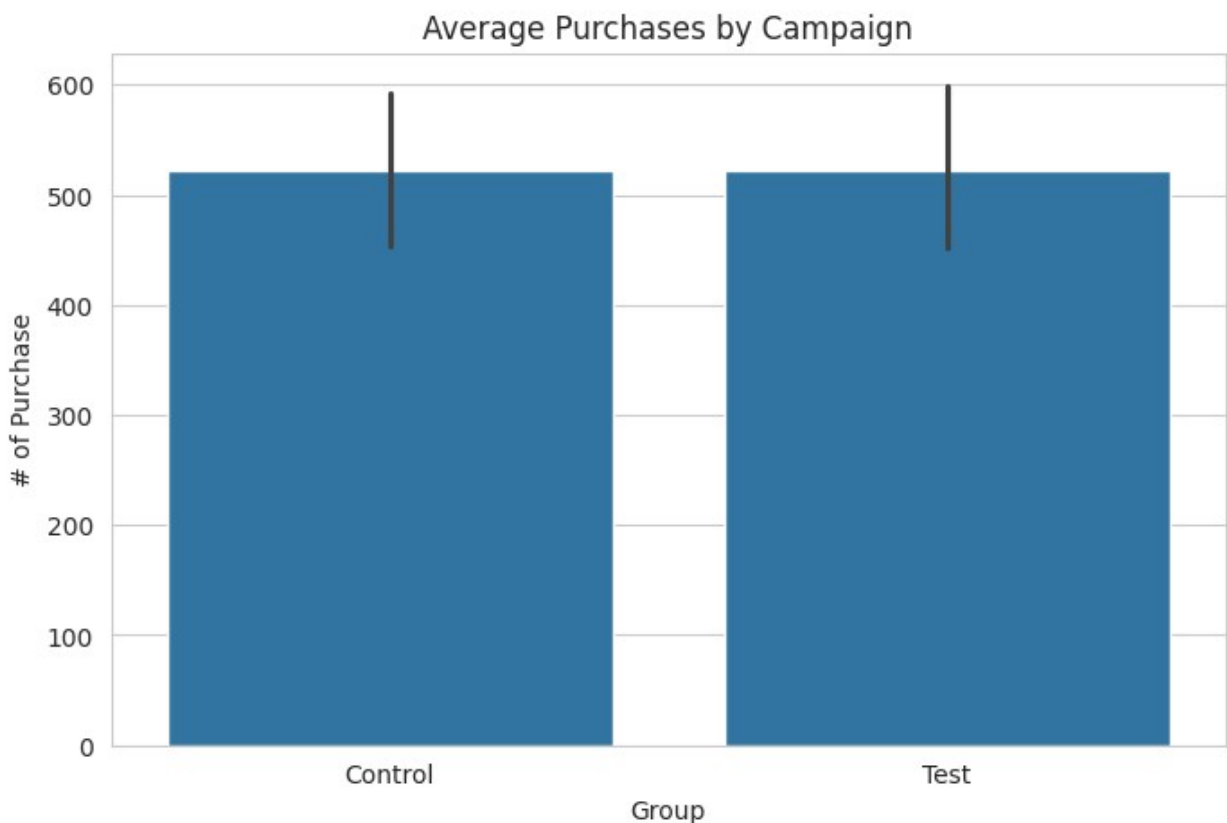
So Control is better at converting traffic into actual purchase behavior.

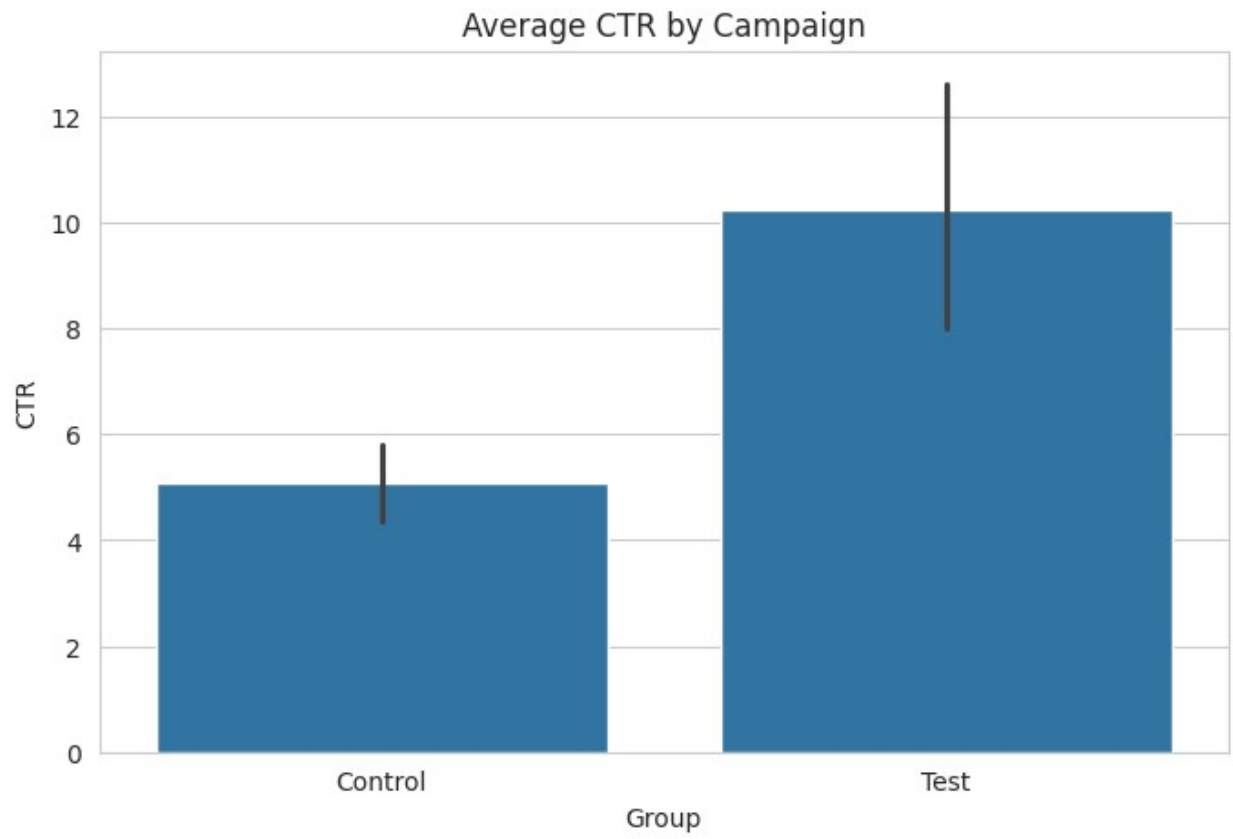
VISUALIZATION OF KPI METRICES

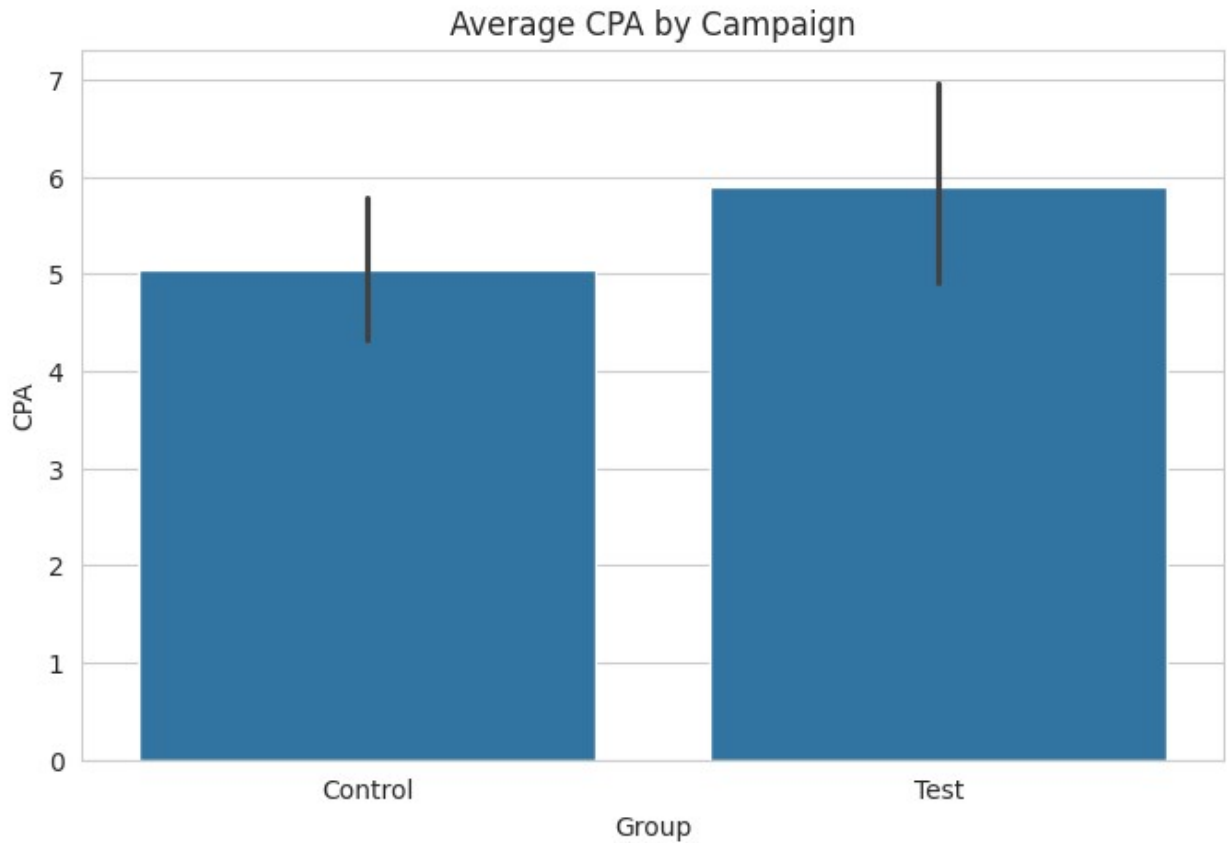
```
plt.figure(figsize=(8,5))
sns.barplot(data=ab_data, x='Group', y='# of Purchase',
            estimator=np.mean)
plt.title('Average Purchases by Campaign')
plt.show()

plt.figure(figsize=(8,5))
sns.barplot(data=ab_data, x='Group', y='CTR', estimator=np.mean)
plt.title('Average CTR by Campaign')
plt.show()

plt.figure(figsize=(8,5))
sns.barplot(data=ab_data, x='Group', y='CPA', estimator=np.mean)
plt.title('Average CPA by Campaign')
plt.show()
```

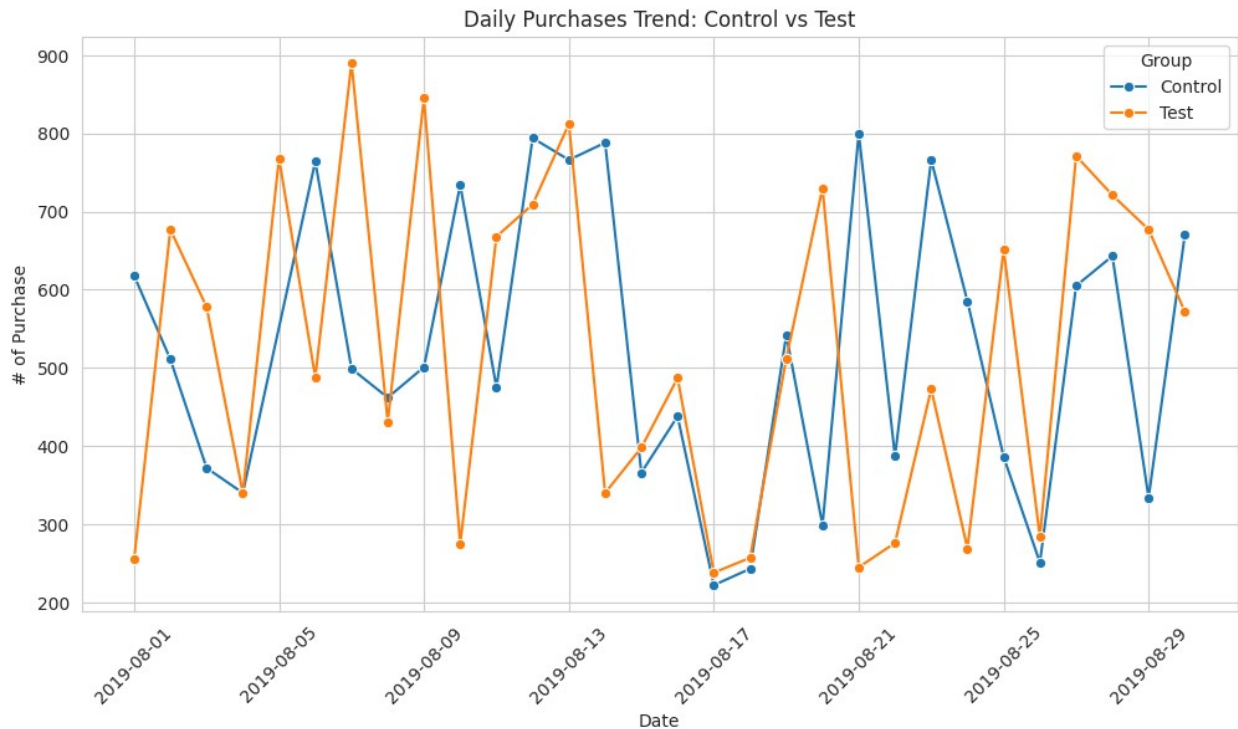






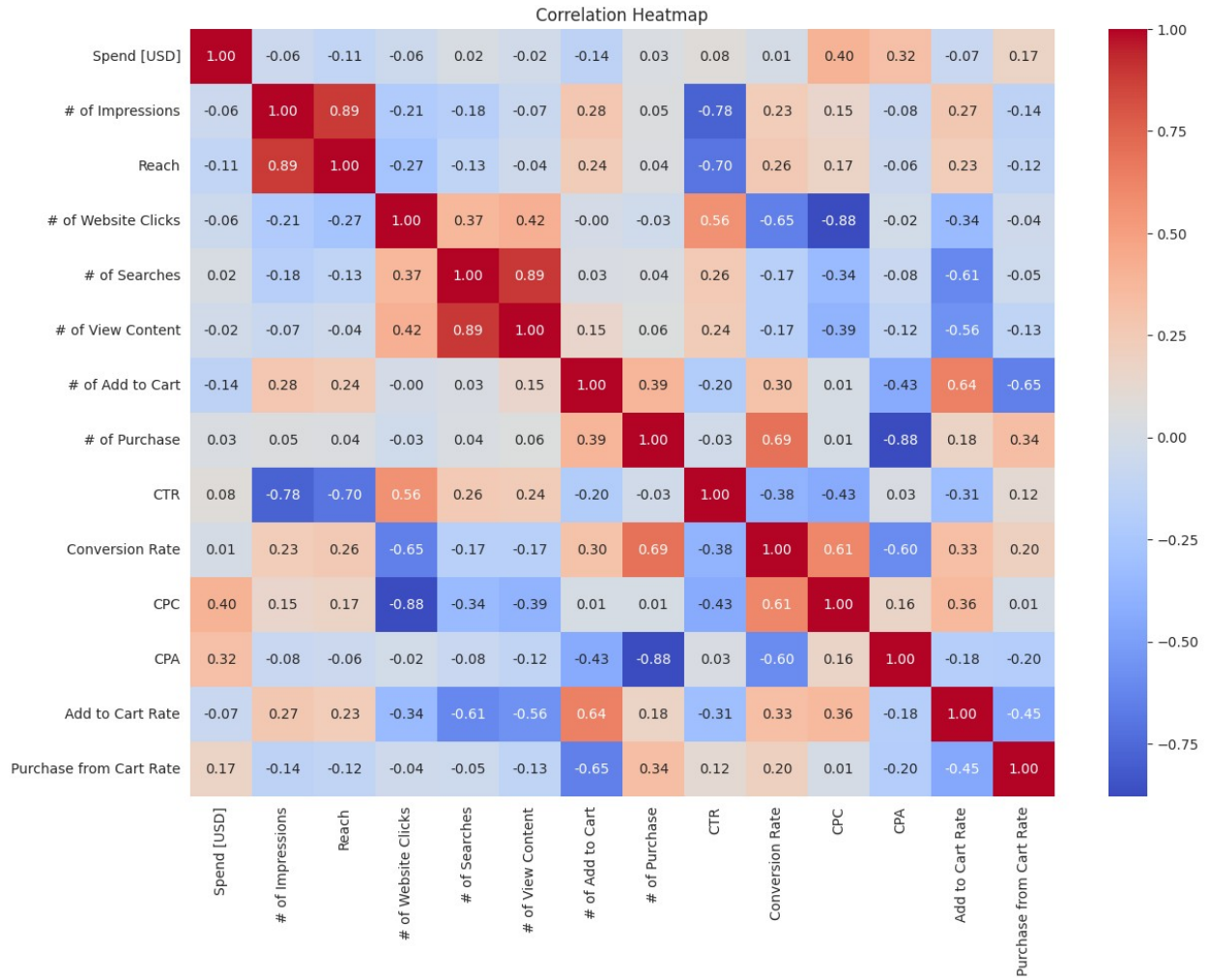
## Daily Purchases Trend: Control vs Test

```
plt.figure(figsize=(12,6))
sns.lineplot(data=ab_data, x='Date', y='# of Purchase', hue='Group',
marker='o')
plt.title('Daily Purchases Trend: Control vs Test')
plt.xticks(rotation=45)
plt.show()
```



## Correlation heatmap

```
corr_data = ab_data.drop(columns=['Campaign Name', 'Date', 'Group'])
plt.figure(figsize=(14,10))
sns.heatmap(corr_data.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap')
plt.show()
```



# Statistical A/B Testing — Hypothesis Testing

We use an Independent Sample T-Test to check whether the difference between the two campaigns is statistically significant.

## Hypothesis for Purchases

### Null Hypothesis ( $H_0$ )

There is no significant difference in purchases between the Control and Test campaigns.

$H_0: \mu_{\text{control}}$  is equal to  $\mu_{\text{test}}$

### Alternative Hypothesis ( $H_1$ )

There is a significant difference in purchases between the two campaigns.

$H_1: \mu_{\text{control}}$  is not equal to  $\mu_{\text{test}}$

## T-Test Code for Purchases

```
control_purchases = ab_data[ab_data['Group'] == 'Control']['# of Purchase']
test_purchases = ab_data[ab_data['Group'] == 'Test']['# of Purchase']

t_stat, p_value = ttest_ind(control_purchases, test_purchases,
                             equal_var=False)

print("T-statistic:", t_stat)
print("P-value:", p_value)

T-statistic: 0.030212884995111548
P-value: 0.9760037958073526

alpha = 0.05
```

```
if p_value < alpha:  
    print("Reject the null hypothesis: Significant difference in  
purchases between Control and Test.")  
else:  
    print("Fail to reject the null hypothesis: No significant  
difference in purchases.")
```

Fail to reject the null hypothesis: No significant difference in purchases.

## Interpretation of Purchase Test

Since:

$$p=0.9760>0.05$$

we fail to reject the null hypothesis.

## Conclusion

There is no statistically significant difference in purchases between the Control and Test campaigns.

## Business meaning

Although the Control campaign has slightly higher average purchases (522.79 vs 521.23), that difference is so small that it can easily be due to random variation.

So from a purchase outcome perspective, the two campaigns are effectively performing the same.

## T-Test Code for other metrics

```
def ab_test_metric(data, metric):  
    control = data[data['Group'] == 'Control'][metric]
```

```

test = data[data['Group'] == 'Test'][metric]

t_stat, p_value = ttest_ind(control, test, equal_var=False)

print(f"\nMetric: {metric}")
print("T-statistic:", round(t_stat, 4))
print("P-value:", round(p_value, 4))

if p_value < 0.05:
    print("Result: Significant difference between Control and
Test")
else:
    print("Result: No statistically significant difference")

metrics = [
    '# of Website Clicks',
    '# of Add to Cart',
    '# of Purchase',
    'CTR',
    'Conversion Rate'
]

for metric in metrics:
    ab_test_metric(ab_data, metric)

```

```

Metric: # of Website Clicks
T-statistic: -1.5761
P-value: 0.1205
Result: No statistically significant difference

```

```

Metric: # of Add to Cart
T-statistic: 4.2375
P-value: 0.0001
Result: Significant difference between Control and Test

```

```

Metric: # of Purchase
T-statistic: 0.0302
P-value: 0.976
Result: No statistically significant difference

```

```

Metric: CTR
T-statistic: -3.9786
P-value: 0.0003
Result: Significant difference between Control and Test

```

```

Metric: Conversion Rate
T-statistic: 1.49
P-value: 0.1428
Result: No statistically significant difference

```



## A) Website Clicks

T-statistic = -1.5761

P-value = 0.1205

Interpretation

Since  $0.1205 > 0.05$ , the difference in clicks is not statistically significant.

Insight

Even though Test has more average clicks, we cannot say with statistical confidence that Test truly outperforms Control in clicks.

## B) Add to Cart

T-statistic = 4.2375

P-value = 0.0000869

Interpretation

Since  $p < 0.05$ , the difference is statistically significant.

Insight

The Control campaign's higher Add-to-Cart performance is real and significant, not random.

Business meaning

Control genuinely performs better at pushing users from browsing into cart activity.

## C) Purchases

T-statistic = 0.0302

P-value = 0.9760

Interpretation

No significant difference.

## D) CTR

T-statistic = -3.9786

P-value = 0.0003388

Interpretation

Since  $p < 0.05$ , the difference in CTR is statistically significant.

Insight

The Test campaign's higher CTR is real and meaningful.

Business meaning

##Test is genuinely better at driving clicks from impressions.

## E) Conversion Rate

T-statistic = 1.4900

P-value = 0.1428

### Interpretation

Since  $p > 0.05$ , the difference in conversion rate is not statistically significant.

### Business meaning

Even though Control's conversion rate looks better numerically, we do not have enough statistical evidence to say it is truly better.

## What the Test Campaign is Better At

### Test campaign strengths

1-Higher CTR (10.24% vs 5.10%)

2-More average website clicks

3-More searches

4-Slightly lower CPC

5-Higher purchase-from-cart rate

### Statistically significant advantage

CTR is significantly higher in Test campaign

### Meaning

The Test campaign is stronger at attracting user attention and generating engagement.

## What the Control Campaign is Better At

### Control campaign strengths

- 1-Higher impressions
- 2-Higher reach
- 3-Higher conversion rate
- 4-Higher Add-to-Cart volume
- 5-Lower CPA
- 6-Slightly more purchases

### Statistically significant advantage

Add-to-Cart performance is significantly higher in Control campaign

### Meaning

The Control campaign is better at driving users deeper into the purchase funnel and doing so at a lower acquisition cost.

## FINAL SUMMARY

```
final_summary = ab_data.groupby('Group')[[  
    'Spend [USD]',  
    '# of Impressions',  
    'Reach',  
    '# of Website Clicks',  
    '# of Searches',  
    '# of View Content',  
    '# of Add to Cart',  
    '# of Purchase',  
]]
```

```

    'CTR',
    'Conversion Rate',
    'CPC',
    'CPA'

```

```

]].mean().round(2)

```

```

print(final_summary)

```

|         | Spend [USD] | # of Impressions | Reach | # of Website Clicks |
|---------|-------------|------------------|-------|---------------------|
| \ Group |             |                  |       |                     |

|         |         |           |          |         |
|---------|---------|-----------|----------|---------|
| Control | 2304.07 | 109559.76 | 88844.93 | 5320.79 |
|---------|---------|-----------|----------|---------|

|      |         |          |          |         |
|------|---------|----------|----------|---------|
| Test | 2563.07 | 74584.80 | 53491.57 | 6032.33 |
|------|---------|----------|----------|---------|

|         | # of Searches | # of View Content | # of Add to Cart | # of Purchase \ Group |
|---------|---------------|-------------------|------------------|-----------------------|
| Control | 2221.31       | 1943.79           | 1300.00          | 522.79                |

|      |         |         |        |        |
|------|---------|---------|--------|--------|
| Test | 2418.97 | 1858.00 | 881.53 | 521.23 |
|------|---------|---------|--------|--------|

|         | CTR   | Conversion Rate | CPC  | CPA  |
|---------|-------|-----------------|------|------|
| Group   |       |                 |      |      |
| Control | 5.10  | 11.48           | 0.49 | 5.05 |
| Test    | 10.24 | 9.23            | 0.47 | 5.90 |

## The Most Important Business Finding

Even though the Test campaign is much better at getting clicks, it does not translate into more purchases.

That is the key takeaway of this project.

In simple words:

1-Test campaign wins at engagement

2-Control campaign wins at shopping intent / purchase efficiency

3-Final purchases are basically the same

## Final Business Conclusion

### Final Conclusion

The Test campaign significantly improves CTR, which means it is better at attracting users and generating clicks.

However, the Control campaign performs better in lower-funnel metrics, especially Add-to-Cart rate and CPA, and it achieves almost the same number of purchases at a lower cost.

Therefore:

If the business goal is brand engagement / traffic / clicks, the Test campaign is better

If the business goal is cost-efficient purchases and stronger funnel quality, the Control campaign is better

## Recommendation

## Recommended decision

If the objective of the campaign is final sales / purchase efficiency, I would recommend continuing with the Control campaign or using the Control campaign as the base strategy.

Why?

Because:

Purchases are statistically the same

Control has lower CPA

Control has higher Add-to-Cart performance

Control seems to bring more purchase-intent traffic

## Strategic Recommendation for the Company

The best practical business recommendation is:

- 1) Use a hybrid strategy
- 2) Use Test campaign creatives/messaging because they clearly improve CTR
- 3) Use Control campaign targeting or landing flow because it performs better in add-to-cart and cost efficiency

This can create a stronger next version of the campaign:

- 1) Better engagement from Test

2) Better purchase funnel performance from Control